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NEUROINSIGHT AI RADIOLOGY REPORT

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Scan ID   : MRI-2026-52947

Date/Time : 29 May 2026 | 12:39 PM

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1. MRI VALIDATION

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Result:

Valid Brain MRI Scan

Validation Confidence:

99.98%

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MRI IMAGE QUALITY ANALYSIS

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Image Resolution:

256x256

Contrast Quality:

Normal

Noise Level:

Low

Artifacts:

No significant imaging artifacts detected.

Image Assessment:

MRI quality is sufficient for AI-based analysis.

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2. TUMOR CLASSIFICATION

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Detection Result:

Tumor Detected

Tumor Type:

Meningioma

Model Confidence:

92.80%

Confidence Interpretation:

High Confidence

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3. TUMOR SEGMENTATION ANALYSIS

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Affected Brain Region:

1.81%

Tumor Area:

1187 pixels

Severity:

Very Small

Location:

[145.24852569502949, 117.34962089300758]

Bounding Box:

[43, 35]

SHAPE ANALYSIS

ECCENTRICITY:

0.584

Description:

Mildly elongated

Interpretation:

Moderate eccentricity

SOLIDITY:

0.963

Description:

Compact and convex

Interpretation:

High solidity - well-defined margin

Shape Risk:

LOW SHAPE RISK - Compact regular morphology.

Segmentation Observation:

A small segmented lesion occupying 1.81% of the brain region was identified. The lesion appears com pact and convex with mildly elongated morphology located near X=145.2, Y=117.3.

TUMOR BEHAVIOR ANALYSIS

Growth Pattern:

Localized and compact

Morphological Nature:

Intermediate morphology

Boundary Characteristics:

Well-defined tumor margins

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EXPLAINABLE AI ANALYSIS

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Grad-CAM Visualization:

Activated

Observation:

The AI model focused primarily on abnormal
high-intensity lesion regions during prediction.

Interpretability Assessment:
The highlighted activation regions are consistent
with segmented tumor localization.

Clinical Relevance:
Grad-CAM improves transparency by explaining
which MRI regions influenced the AI prediction.

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4. TUMOR PROFILE

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Description:
Usually benign extra-axial brain tumor arising from meninges.

Characteristics:

- Well-defined margins
- Slow-growing
- Extra-axial lesion

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5. RISK ASSESSMENT

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Risk Level:
LOW RISK

Clinical Severity Interpretation:
Minimal abnormality detected

Suggested Action:
Regular MRI monitoring recommended.

Prognostic Indicator:
Favorable prognosis likely with monitoring

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6. AI MEDICAL SUMMARY

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MRI findings suggest meningioma with localized tumor involvement.

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FINAL RADIOLOGY IMPRESSION

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Findings are suggestive of Meningioma with approximately 1.81% regional involvement.

Overall Assessment:

Routine monitoring advised.

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7. RECOMMENDATIONS

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1. Neurology consultation recommended
2. Periodic MRI follow-up advised
3. Monitor tumor growth progression

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AI TECHNICAL ANALYSIS

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Primary Classification Model:

ResNet50 CNN

Segmentation Architecture:

U-Net Deep Learning Network

Inference Pipeline:

MRI Validation → Tumor Classification → Segmentation

AI Decision Support:

Enabled

Explainable AI:

Grad-CAM Integrated

Report Generation:

Automated AI-assisted radiology workflow

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SYSTEM RELIABILITY

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Classification Confidence:
92.80%

Segmentation Reliability:
High

AI Consistency:
Prediction confidence and segmentation findings
show strong agreement.

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8. DISCLAIMER

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This report is AI-generated and intended for
educational/research purposes only.

Final diagnosis must always be confirmed
by qualified medical professionals.

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END OF REPORT

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